

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain the Bernoulli's theorem with the help of diagram and illustrate its applications. (CO3)
- Q.24 Differentiate between Otto cycle and Diesel cycle. (CO2)
- Q.25 Explain the construction and Working of Centrifugal Pump with the help of neat sketch. (CO3)

No. of Printed Pages : 4

220333/180332/170345

Roll No.

/030331

3rd Sem / Automobile

Subject : Basics of Thermodynamic, hydraulics and Pneumatics

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Otto cycle is known as - (CO3)
- a) Constant pressure cycle
 - b) Constant Temperature cycle
 - c) Constant entropy cycle
 - d) Constant volume cycle
- Q.2 A pump is a device which converts _____. (CO2)
- a) Mechanical energy into hydraulic energy
 - b) Electric energy into mechanical energy
 - c) Hydraulic energy into mechanical energy
 - d) All of above
- Q.3 Hydraulic ram works on the principle of (CO2)
- a) Water hammer
 - b) Centrifugal action
 - c) Reciprocating action
 - d) All of the above

(300)

(4) 220333/180332/170345
/030331

(1) 220333/180332/170345
/030331

Q.4 A system which consists of single phase is known as (CO1)

- a) Heterogeneous system
- b) Open system
- c) Closed system
- d) Homogeneous system

Q.5 First law of thermodynamics deals with (CO1)

- a) conservation of energy
- b) conservation of mass
- c) Conservation of force
- d) Conservation of momentum

Q.6 Surface tension of a liquid (CO3)

- a) Increases with area
- b) Decreases with temperature
- c) Increases with temperature
- d) Decreases with area

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define universe. (CO1)

Q.8 What do you mean by viscosity. (CO2)

Q.9 Name three types of rotary pumps. (CO2)

Q.10 An envelope which separates the system from surroundings is called as _____ (CO2)

Q.11 Specific volume is the reciprocal of mass density.(True/False) (CO3)

Q.12 SI unit of Surface tension is Newton per metre. (True/False) (CO3)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Differentiate between Open and Closed system. (CO2)

Q.14 Explain Bourdon tube pressure gauge with the help of neat sketch. (CO2)

Q.15 Explain Bernoulli's theorem. (CO1)

Q.16 Explain steady flow and unsteady flow. (CO2)

Q.17 What are differential manometers? (CO2)

Q.18 Write down any four advantages of Pneumatic system. (CO1)

Q.19 Write steady flow energy equation for Boiler and Nozzle. (CO3)

Q.20 Differentiate between homogeneous and heterogeneous system. (CO3)

Q.21 Explain the vane pump with the help of neat sketch. (CO1)

Q.22 Differentiate the uniform and non uniform flow. (CO2)